

Fall 2026 Semester
Cryptographic Application



Lab Environment Basics

LIBERTAS
JUSTITIA
VERITAS

Wonsuk Choi
beb0396@korea.ac.kr
School of Cybersecurity
Korea University, Rep. Korea

0. How the Lab Environment Works

- The course repository is opened in VS Code. Docker runs the lab environment.
- A mounted folder connects the two.

```
Your computer
├─ VS Code: read README/PDF and edit code
├─ Docker Desktop
│   └─ Course container
│       └─ /workspace ← course repository is mounted here
```

Key idea: You edit the file once in VS Code. The container sees the saved file through /workspace.



1. What is Docker?

- Docker is a tool for running software in a prepared environment.
- In this course, **Docker provides the same Ubuntu-based lab environment to every student**, even if their own computer is macOS or Windows.

Why we use Docker

- a. Same tools and versions for everyone
- b. Less setup on each student computer
- c. Easier grading and troubleshooting

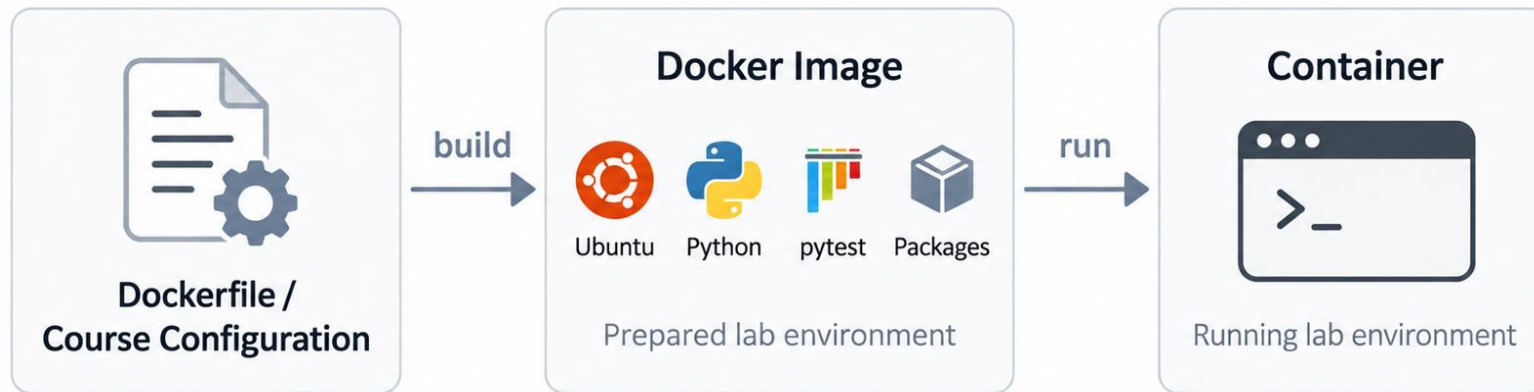
What Docker does not mean

You do not write “Docker code.”
You still write normal Python files for the lab.



2. What is a Container?

- A **container** is a running lab environment created by Docker.
- The course container contains Ubuntu and the tools needed for the labs.
- When you enter the container, your terminal commands run inside this prepared environment.



3. What is a Mount?

- A **mount** makes a folder from your computer visible inside the container.

```
Course repository   on HOST
    ↓ mount
/workspace          inside CONTAINER
```

Because of the mount, the container can run the files that you edited and saved in VS Code.



Image Build vs. Mount

- A **Docker image** contains the prepared environment required for the lab.

```
docker compose -f docker/compose.yml build
```

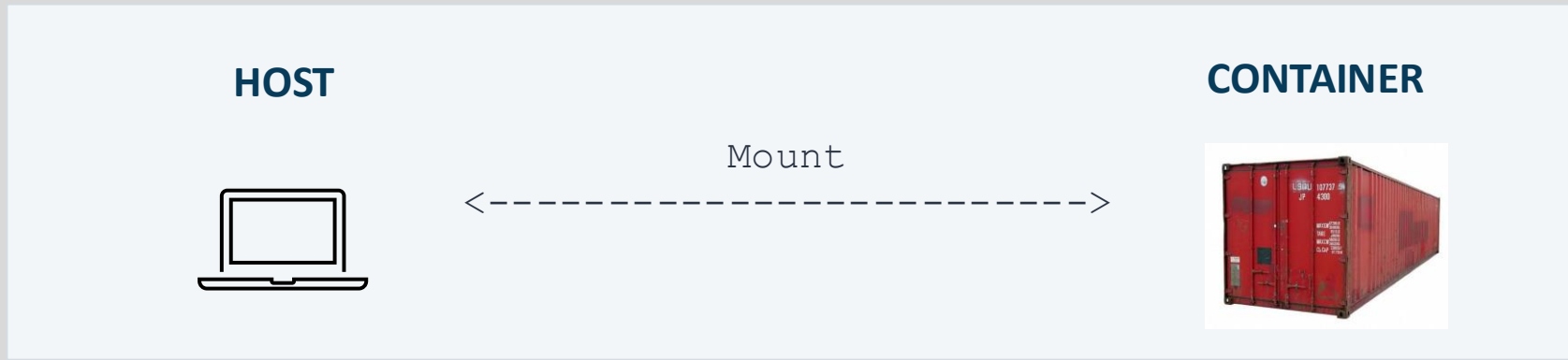
- The image may include:
 - Ubuntu
 - Python
 - pytest
 - Required packages
 - Course tools ...

You do not need to rebuild the image every time you modify your Python code!



Image Build vs. Mount

- A **mount** connects the course repository on the HOST to the container.



- Changes saved under the mounted `/workspace` directory are **immediately** reflected on both the host and inside the container.

4. HOST vs. CONTAINER

HOST

Your computer

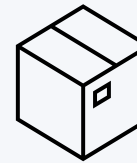
Use HOST to open VS Code, edit files, and start the Docker container.



CONTAINER

Docker lab environment

This is where Python, pytest, and lab commands should run.



5. Where Do I Edit My Code?

- Docker provides the execution environment. The mounted `/workspace` directory stores your work on the host.
- **Edit on the HOST, run inside the CONTAINER.**

Do this

- Open the lab file in VS Code Explorer.
- Edit the Python code.
- Save the file.

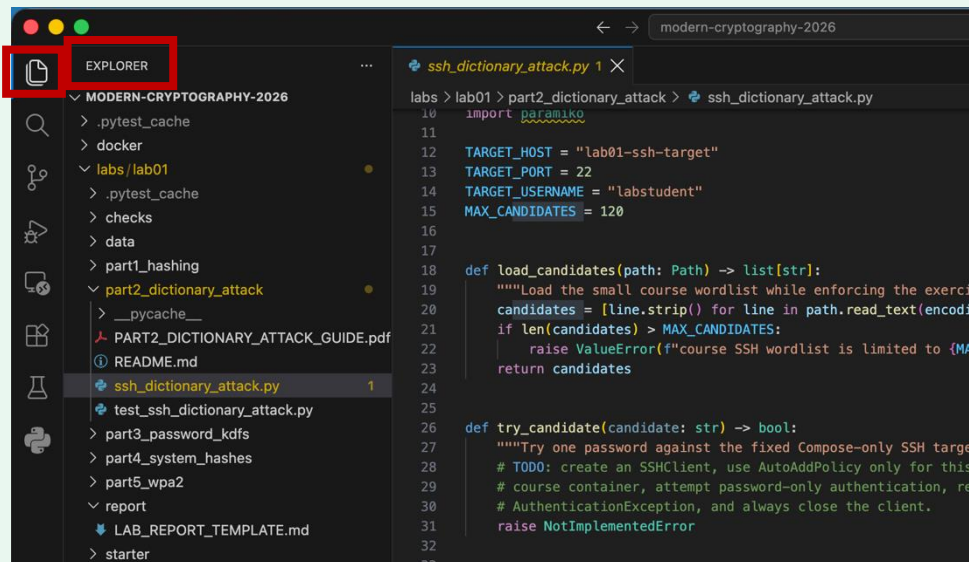
Be careful

- Files under `/workspace` are shared with the host.
- Files outside `/workspace` may be deleted when the temporary container is removed.

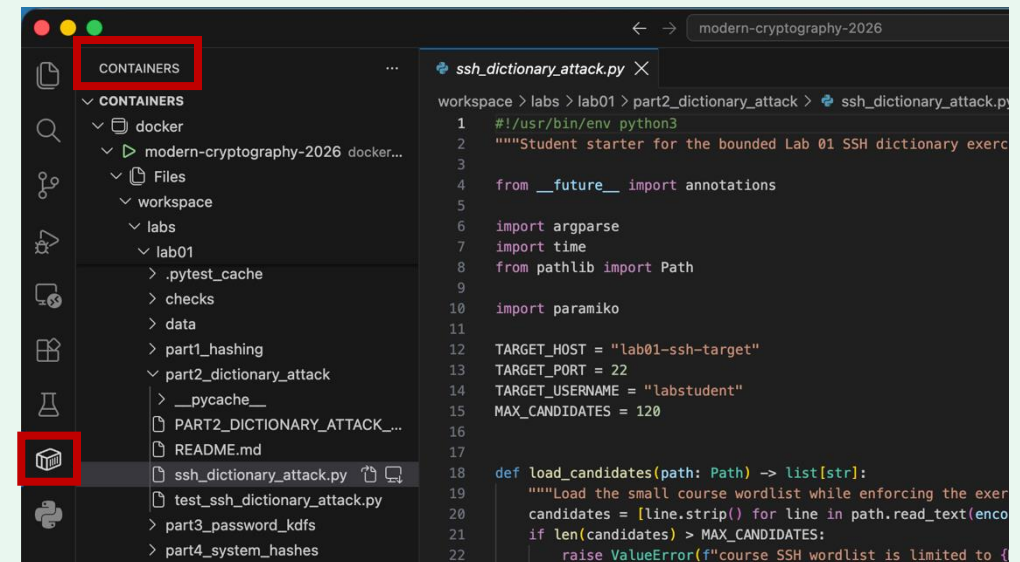
5. Where Do I Edit My Code?

- Files under `/workspace` are preserved because they are stored in the mounted host folder.

Edit the file in *EXPLORER*



Edit the file in *CONTAINERS*



6. Where Do I Run My Code?

- Run lab commands inside the Docker container.

1) Use this for automated checks.

```
cd /workspace/labs/lab01  
pytest -q
```

2) Use this for running one program.

```
cd /workspace/labs/lab01/part1_hashing  
python3 hash_password.py
```

Do not run lab commands directly in macOS Terminal, Windows PowerShell, or the WSL host terminal. Run them inside the course container unless instructed otherwise.

